

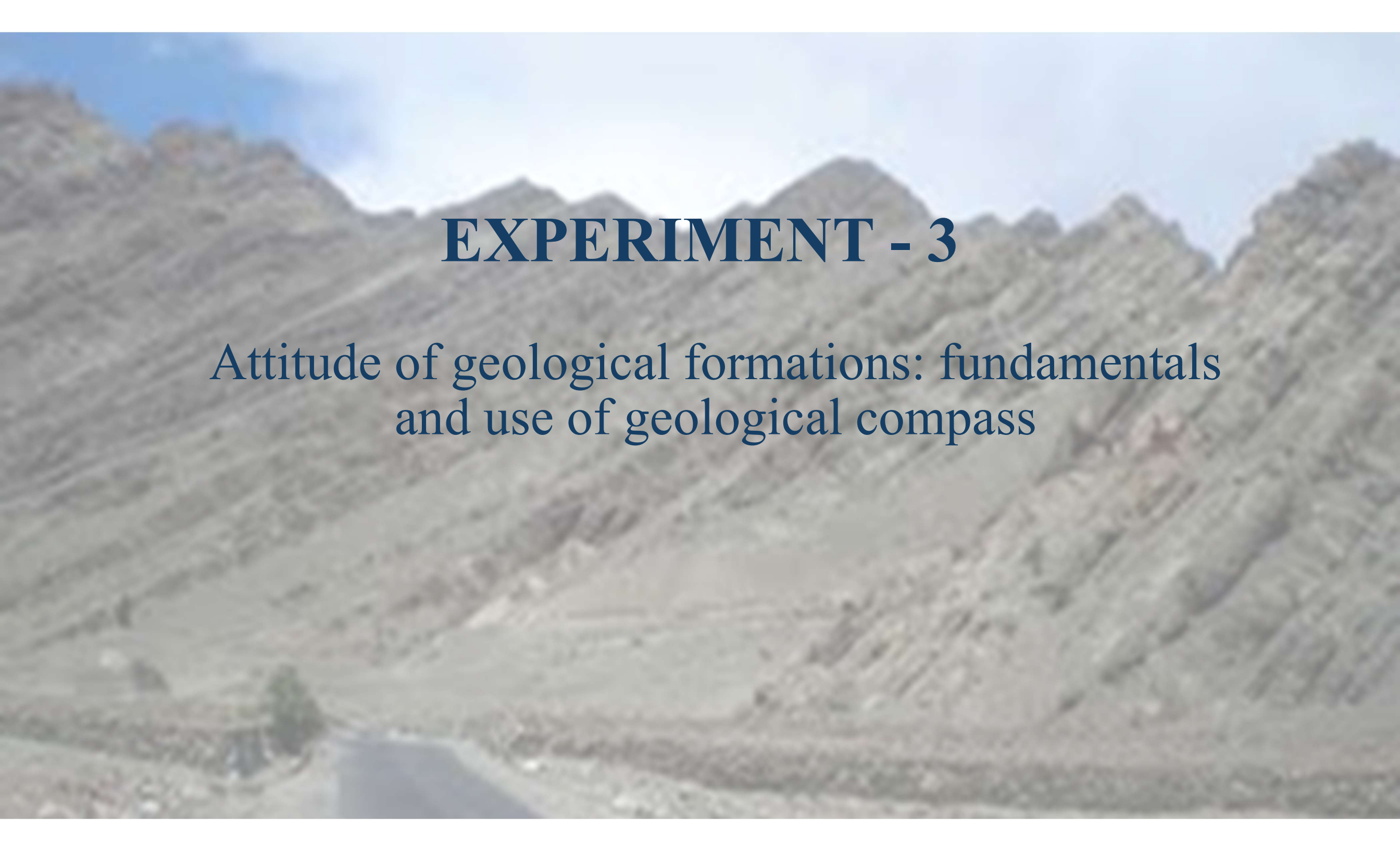
Engineering Geology Lab & Geological Engineering Lab CVP-121 & CVP2401



Course Instructor

Prof. Deepanshu Shirole

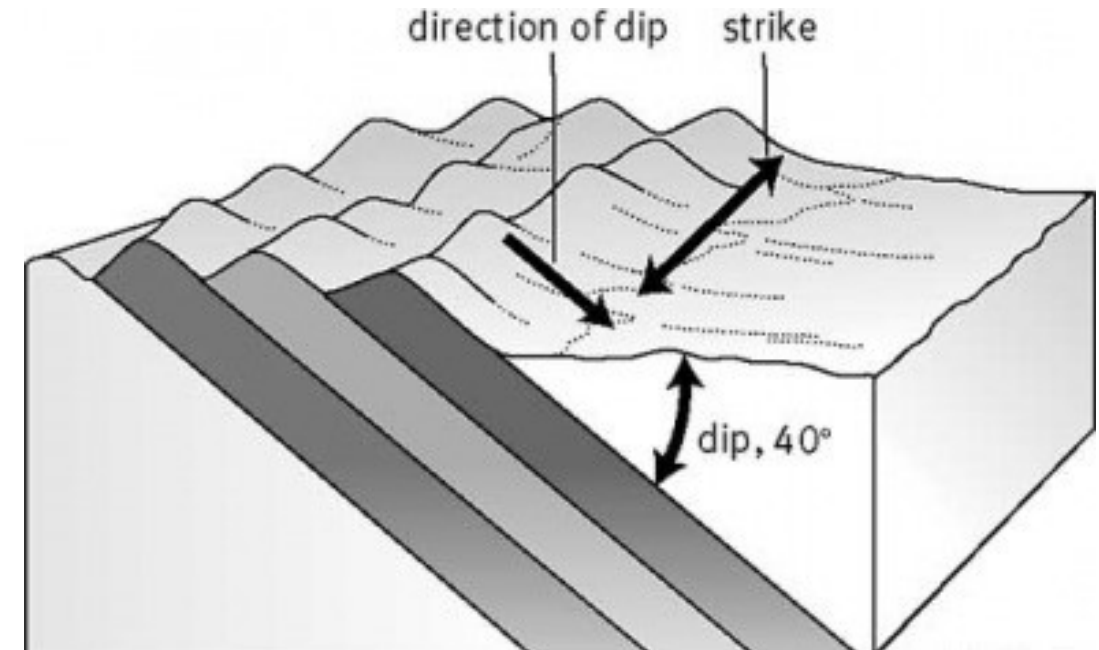
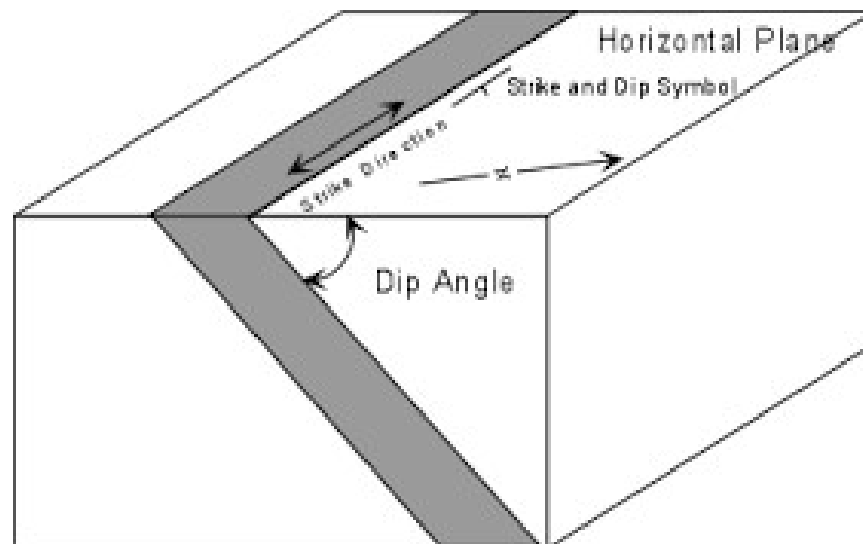
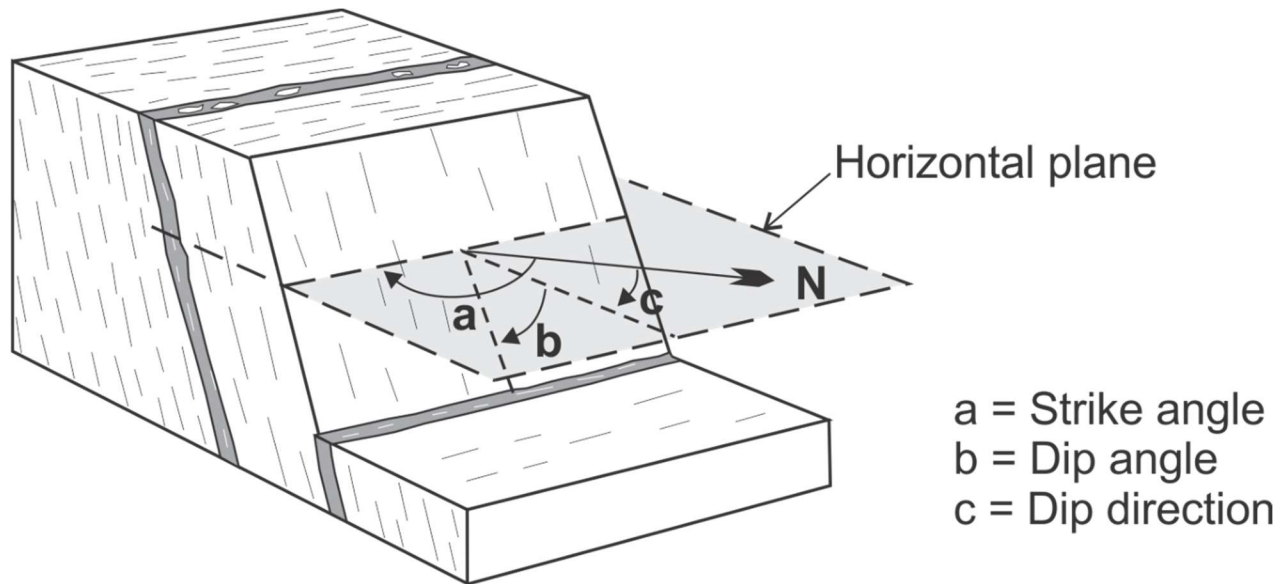
Department of Civil and Environmental
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An aerial photograph of a mountain range. The mountains are covered in dense vegetation, but the underlying geological structures are clearly visible as distinct, tilted layers of different colors (shades of brown, tan, and green). The sky is a pale blue with some light clouds. The text is overlaid on the central part of the image.

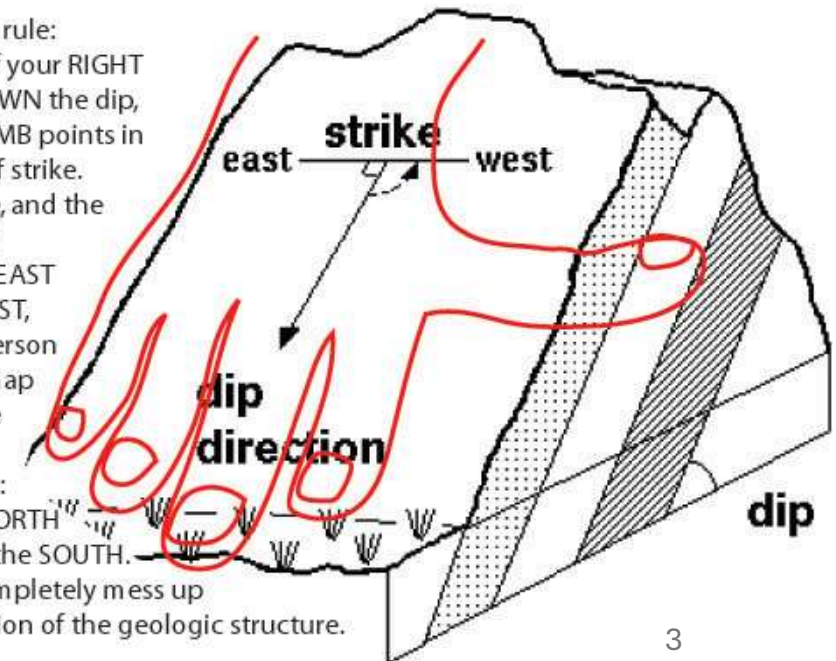
EXPERIMENT - 3

Attitude of geological formations: fundamentals
and use of geological compass

Orientation of planes



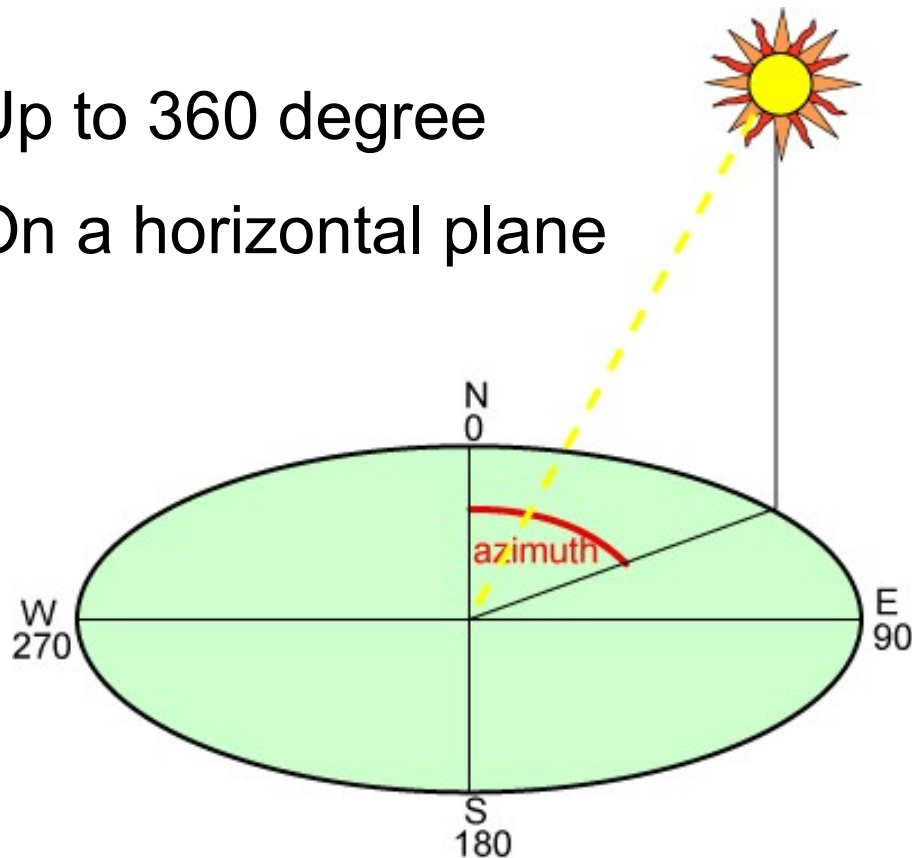
The right hand rule:
If the fingers of your RIGHT hand point DOWN the dip, then your THUMB points in the direction of strike. Ignore this rule, and the strike could be interpreted as EAST rather than WEST, and another person reading your map could think the strata dip the other direction: down to the NORTH rather than to the SOUTH. That would completely mess up the interpretation of the geologic structure.



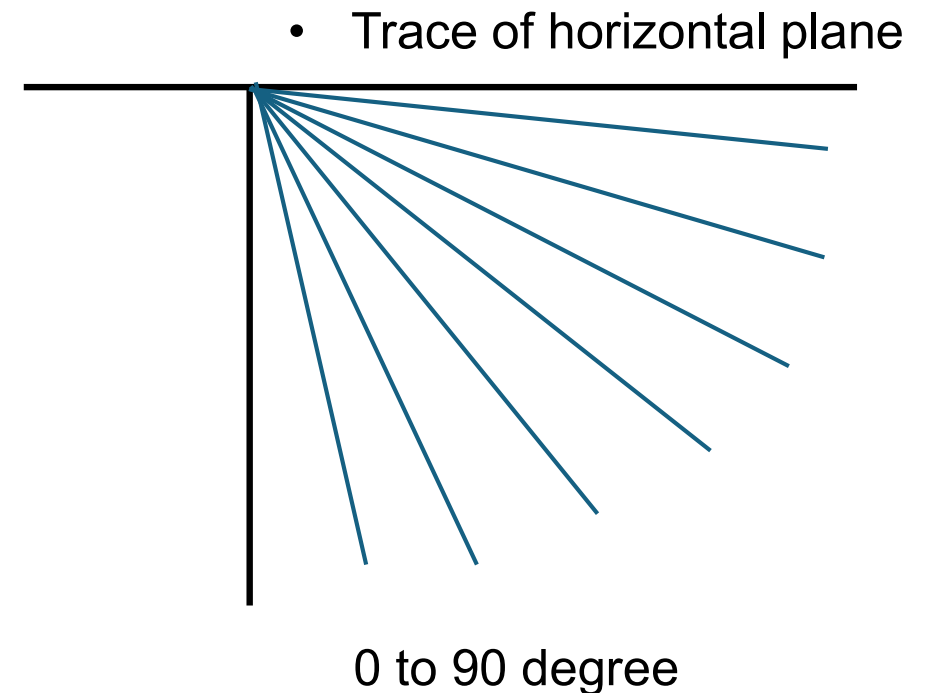
Azimuth (direction) and inclination

- **Azimuth** is measured along horizontal plane in degrees wrt North.

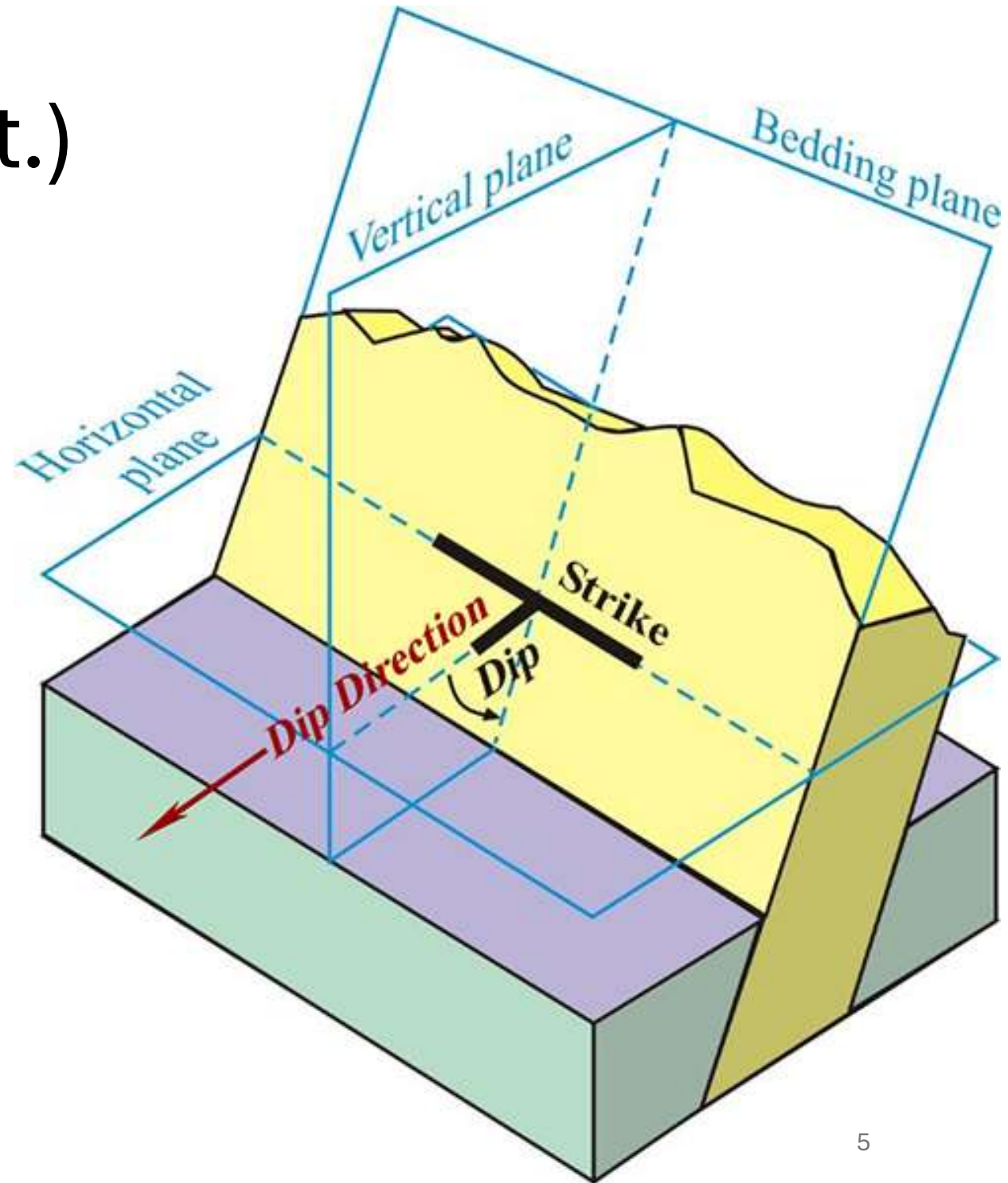
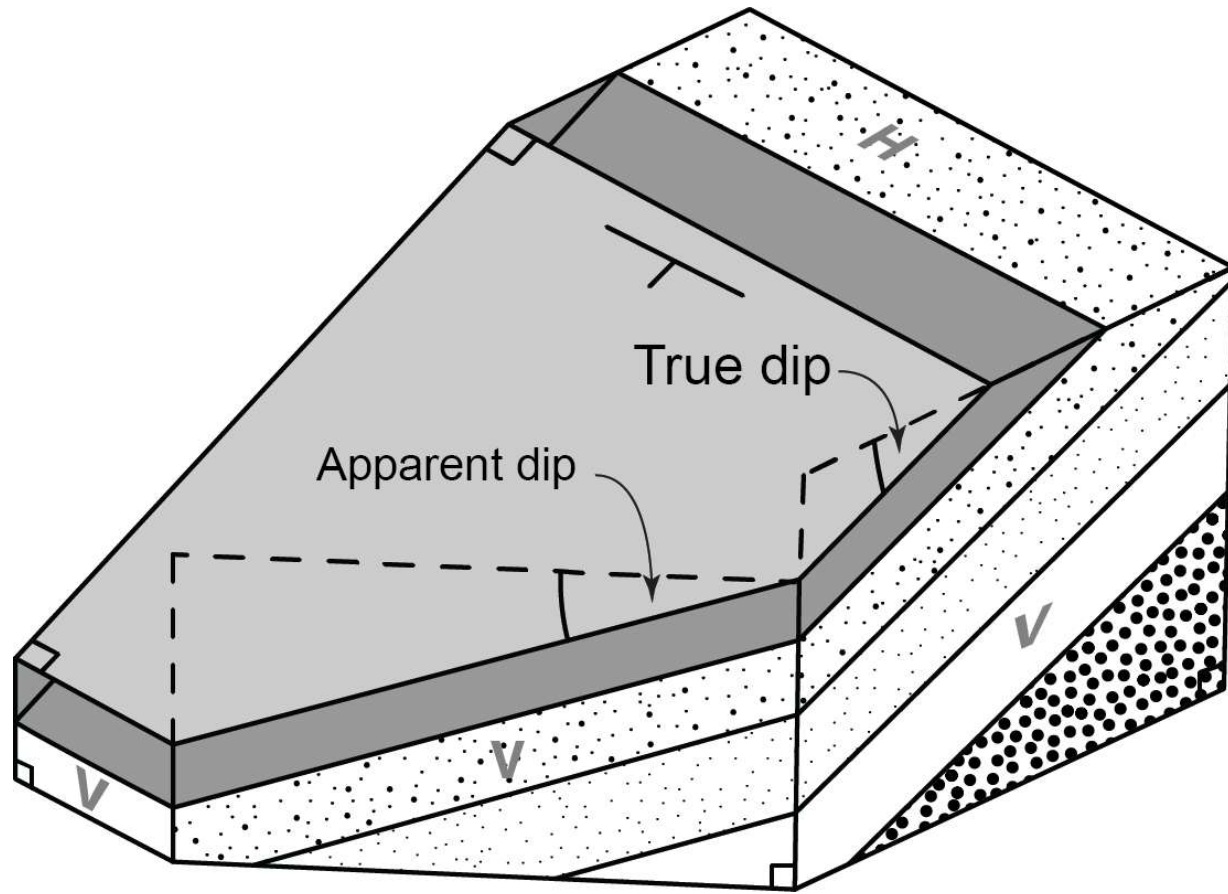
- Up to 360 degree
- On a horizontal plane



- **Inclination** is measured on the vertical plane. It is the angle of tilt/slope at which the plane/line is oriented from the horizontal plane



Orientation of planes (cont.)



Introduction

1. Strike:

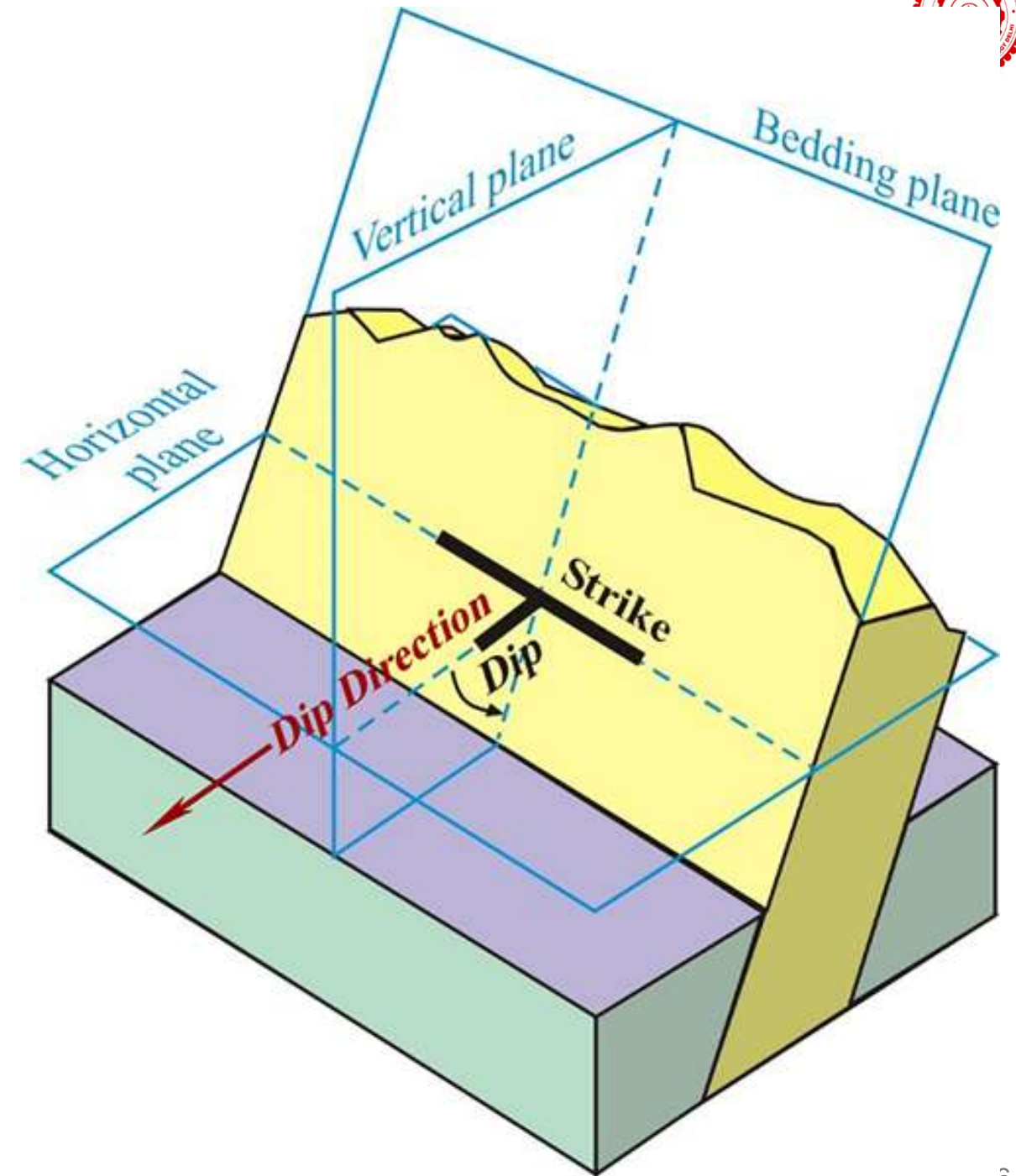
- Direction of the line formed by intersection of bedding plane and horizontal plane.
- It is always measured with reference to true North and South.

2. Dip:

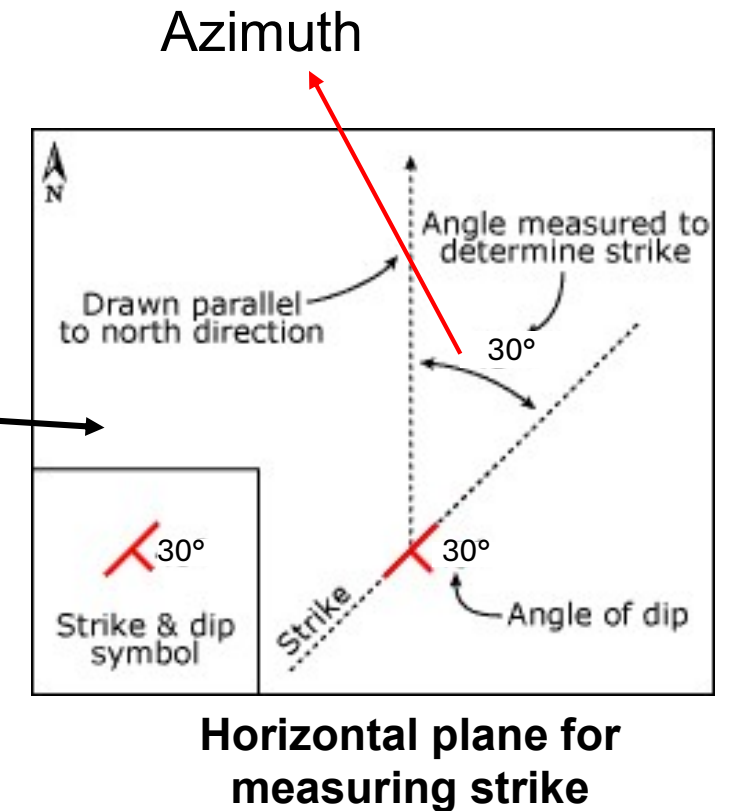
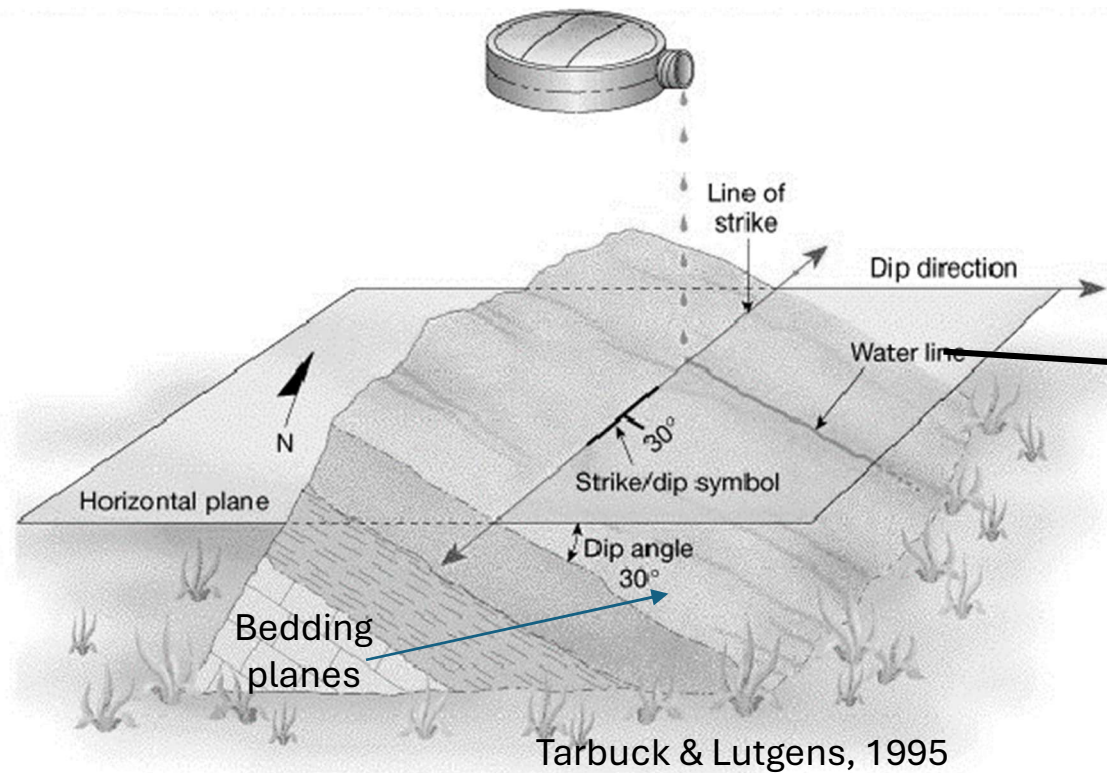
- It is the angle of inclination of bedding plane (Rock bed) with the horizontal plane.

3. Dip Direction:

- The compass direction towards which the plane slopes.



Orientation of planes (cont.)



- **Strike:** angle of the strike line with the north direction, measured in the horizontal plane.
- **Dip:** angle at which a plane is inclined with the horizontal plane. Measured in vertical plane perpendicular to strike. (True vs apparent ?)

Geological Compass



- Specialized tool used by geologists to measure the orientation (or attitude) of geological features.
- **Types of geological compass:**
 - Brunton compass
 - Silva (or Mirror) compass
 - Suunto compass
 - Freiberg compass
 - Digital geological compass



Parts of Compass



- **Directional Arrow:**

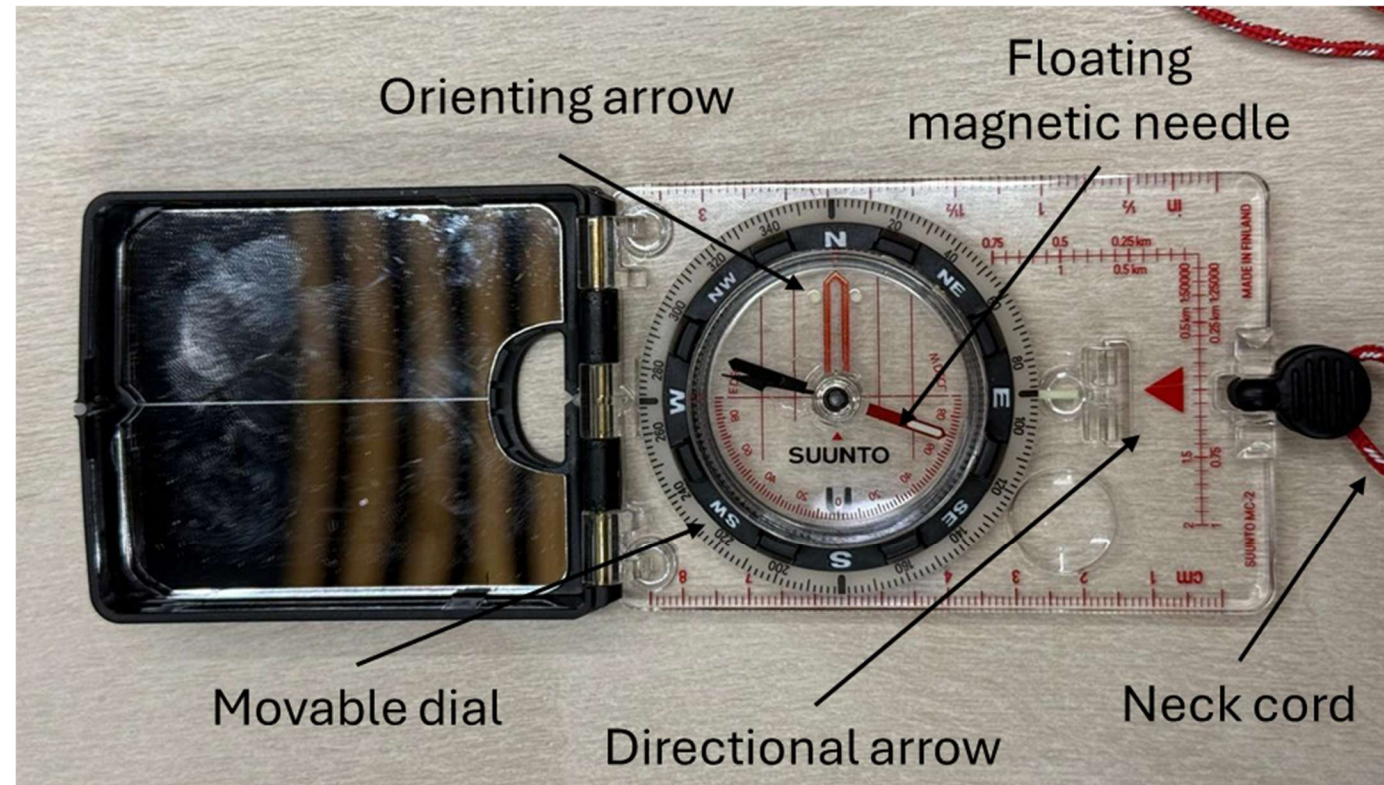
It points **away from you**, showing the direction of travel when taking a bearing or navigating.

- **Orienting Arrow (Rotates with Dial):**

The red outline inside the capsule that rotates when you turn the dial. Used to align with the magnetic needle to set a bearing.

- **Floating Magnetic Needle:**

The red-and-black needle that moves freely inside the liquid-filled housing. The red end always points to magnetic north.



Parts of Compass

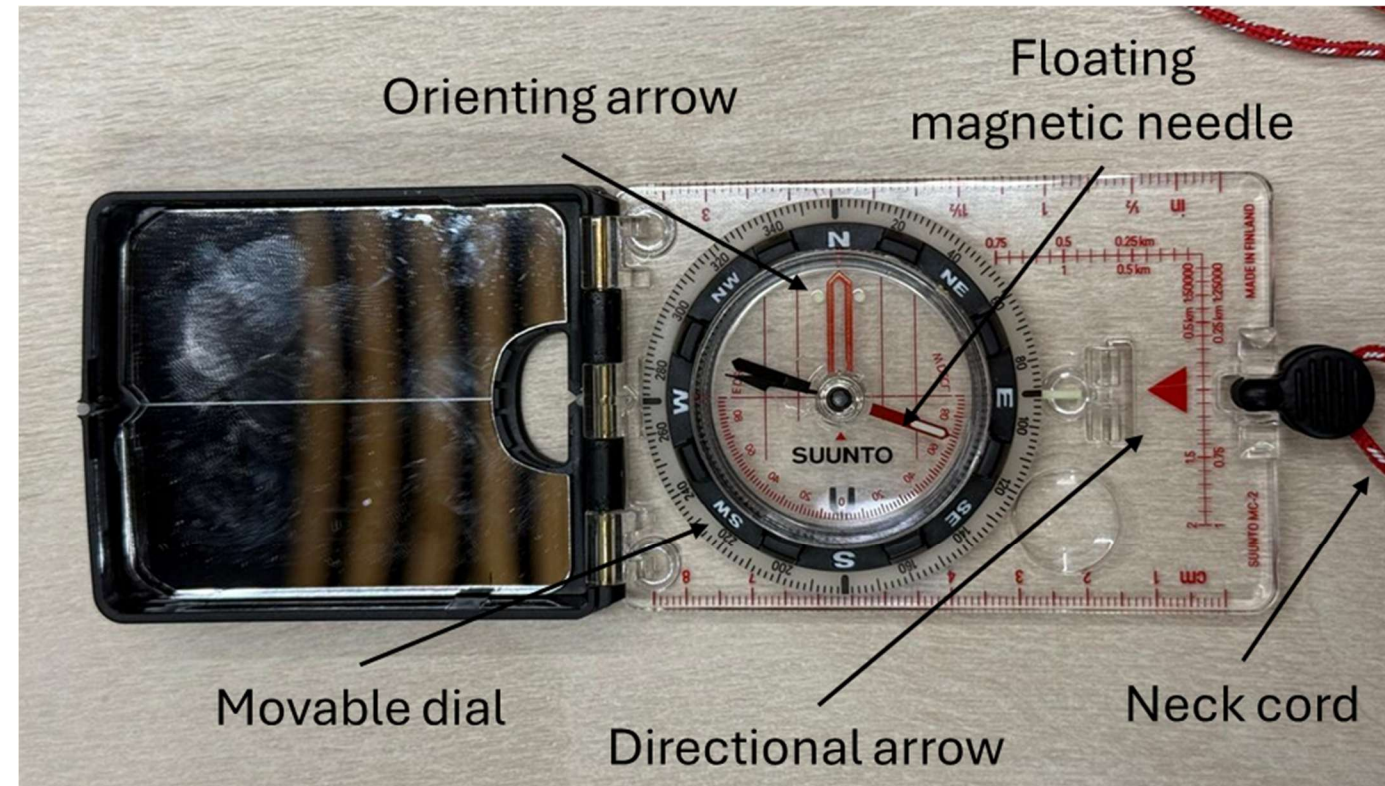


- **Moveable Dial:**

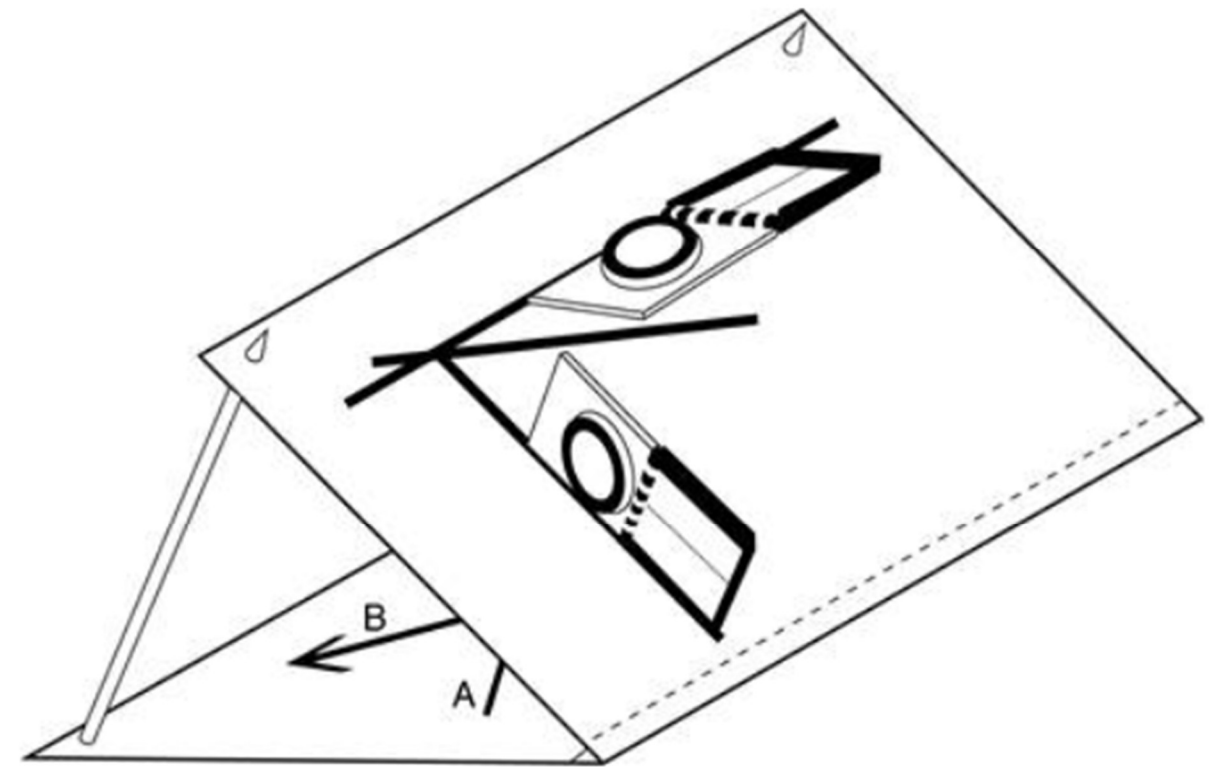
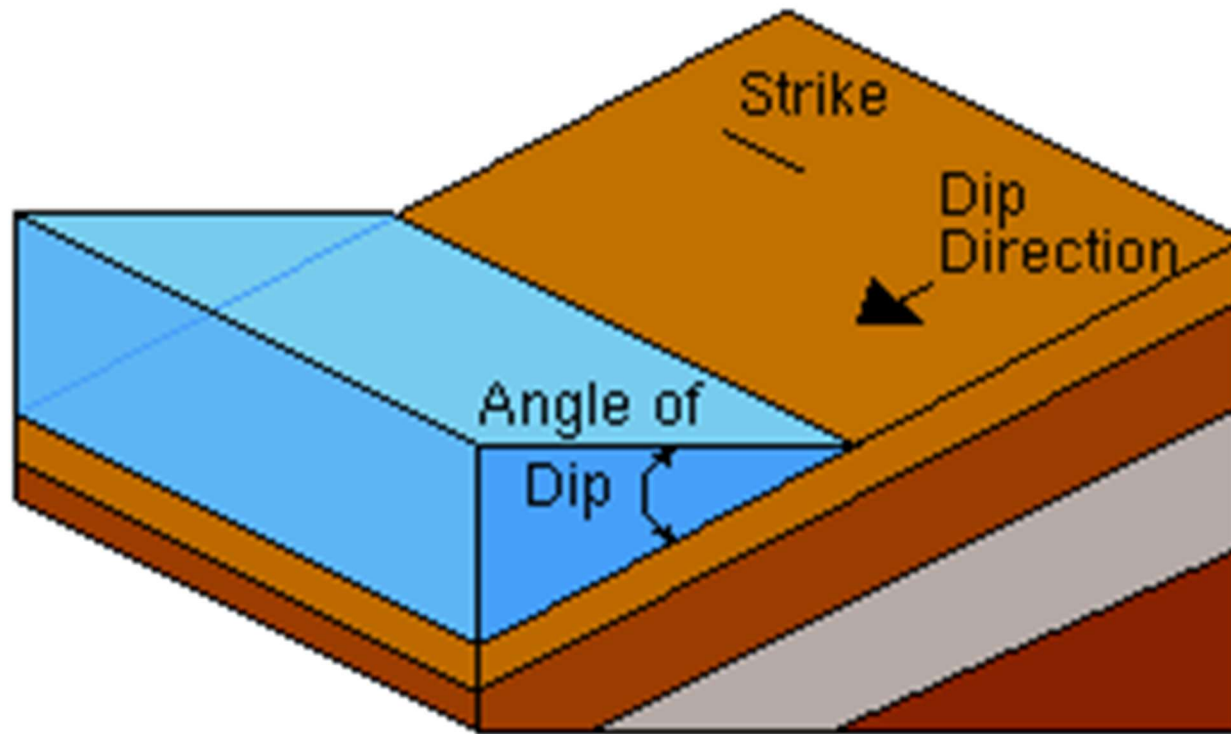
The rotating ring around the compass housing marked in degrees (0–360°). Used to **set bearings** by aligning the orienting arrow with the magnetic needle.

- **Neck Cord:**

A cord attached to the compass for wearing around the neck. Helps prevent losing the compass while in the field.



Strike and Dip measurement:

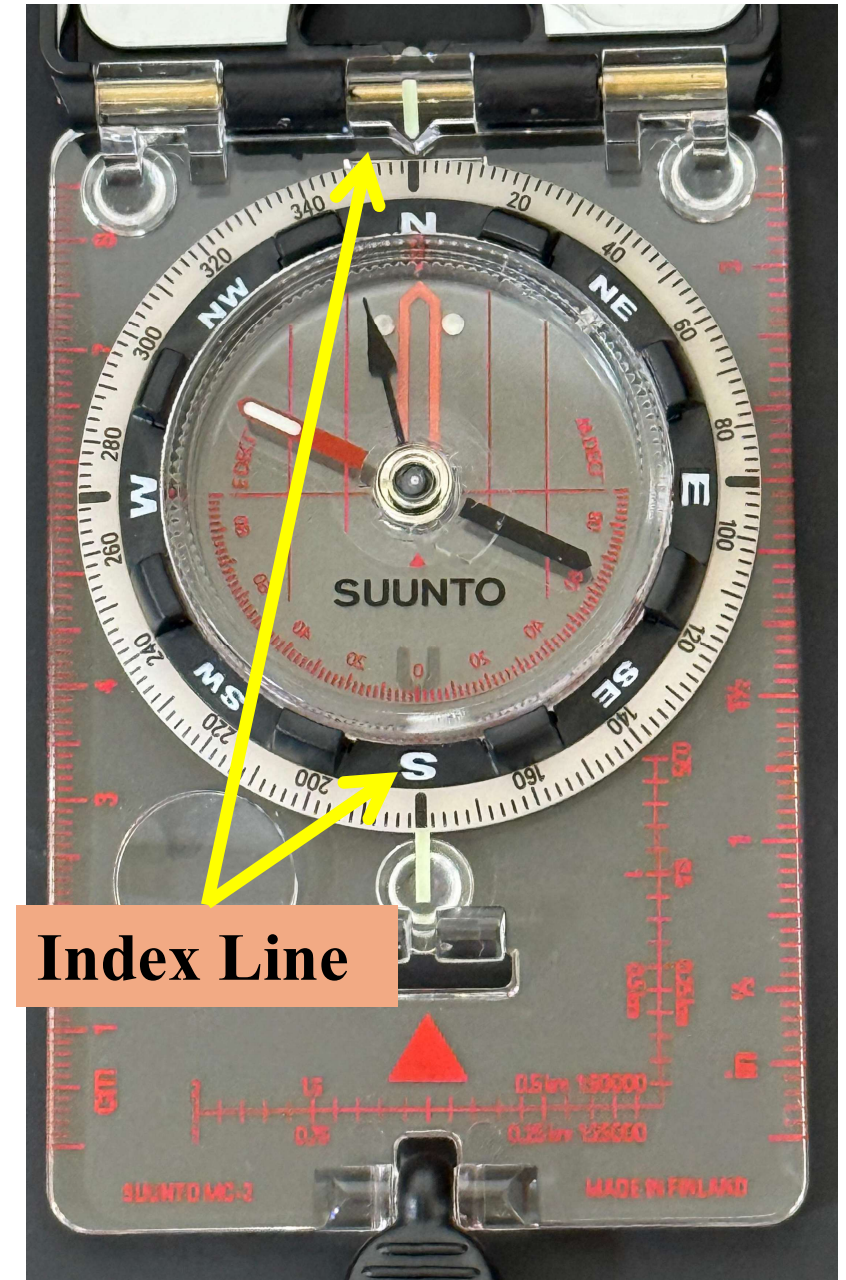


Strike Measurement



Strike measurements in steps

1. Find longitude & Latitude of TARGET LOCATION with the help of google maps.
2. Find a clearly visible rock surface such as a bedding plane.
3. Align the North & South of the movable dial of the compass with the index line.
4. For strike measurement always focus on movable dial (outer readings), floating magnetic needles and orienting arrows.



Strike Measurement



4. Align the compass parallel to the horizontal line on the plane.
5. Ensure full contact between the bedding plane and the compass.
6. Measure the angle pointed by magnetic North needle (Red needle with white mark).



Strike Measurement



7. Now, Move the orienting arrow towards the magnetic North and find the angle and direction with respect to magnetic North.



Dip Measurement



Dip measurements in steps

1. Align the East & West of the compass with the index line.
2. Find a clear planar surface like a bedding plane.
3. For dip angle measurement always focus on inner readings (half circle) and free fall arrow (Black colour arrow with needle).

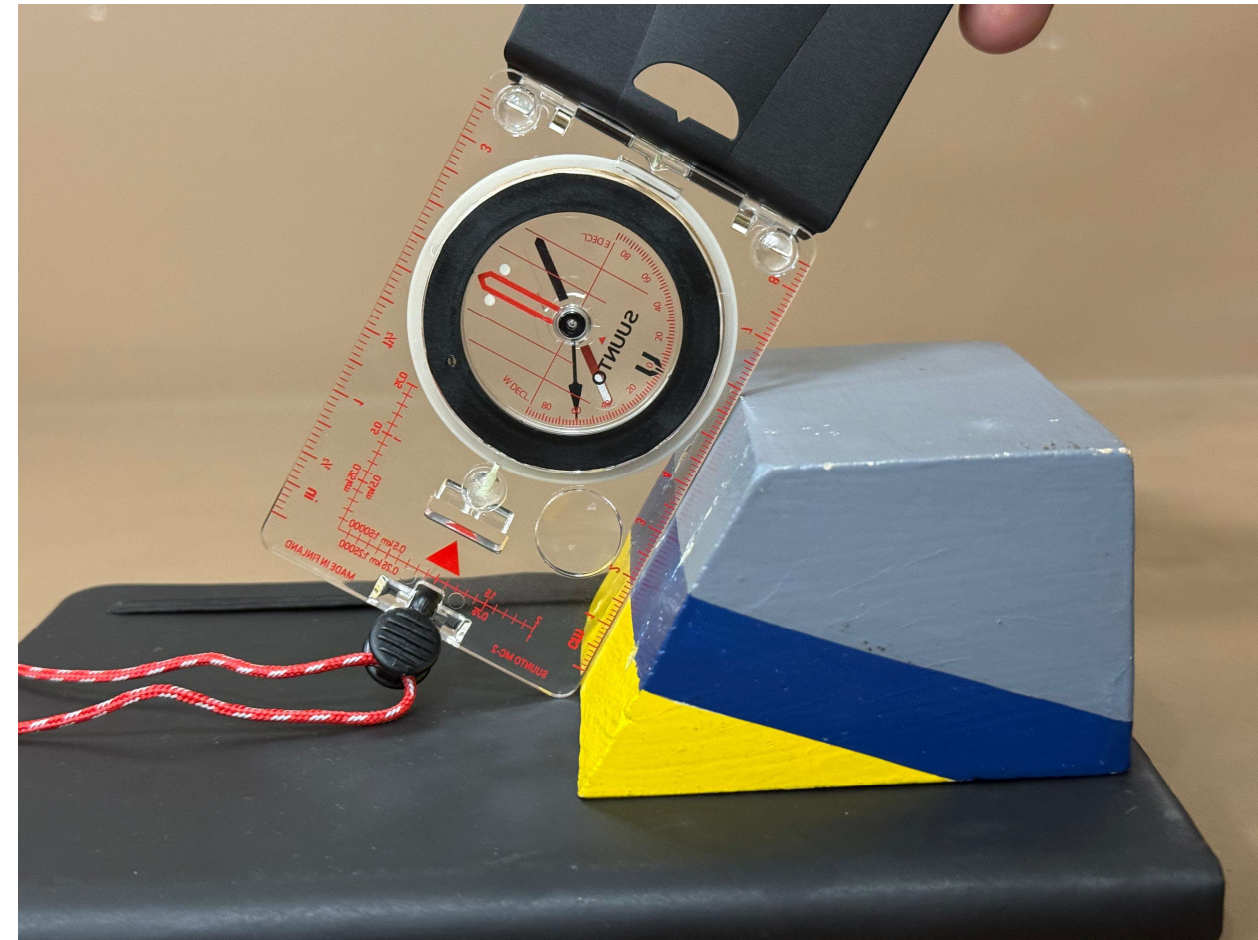


Free Fall Arrow

Dip Measurement



4. Place the long edge of the compass perpendicular to the strike (i.e., in the steepest direction of the plane).
5. Ensure full contact between the bedding plane and the compass.
6. Read the dip angle from the compass (in degrees), this gives the dip angle.



An aerial photograph of a mountain valley. The landscape is a patchwork of green fields, some of which appear to be agricultural, interspersed with dense, dark green forests. The valley is surrounded by rolling hills and mountains in the background, creating a sense of depth. The lighting suggests a bright, sunny day.

Thank You